COLLEGE OF EDUCATION
TEXAS TECH

EPSY 5383: DATA ANALYSIS WITH STATISTICAL SOFTWARE

COURSE INFORMATION

- **Semester:** Spring, 2026
- **Modality:** Face-to-face

INSTRUCTOR INFORMATION

- **Dr. Jaehoon (Jason) Lee**
- **Office:** Education 378
- **Office Hours:** By appointment
- **Phone:** 806.834.5907
- **Email:** jaehoon.lee@ttu.edu

COURSE DESCRIPTION

This course delves into the essential principles and practices of quantitative data analysis using statistical software. It covers a comprehensive range of topics, including:

- Survey construction
- Data collection and visualization
- Intermediate statistics and psychometric tests
- Mediation analysis
- Reporting findings

COURSE LEARNING OUTCOMES

- **Research and Dissemination**: Understand the core concept of quantitative research methods and statistical tests

- **Practical Application**: Gain hands-on exercise in conducting data analysis with statistical software.

- **Research and Dissemination**: Develop skills to design and implement original research and effectively present findings to target audiences.

COURSE WEBSITE: RaiderCanvas

- The course website hosts the latest version of the syllabus, alongside additional and supplemental readings, lecture notes, and other important course materials. This includes announcements, assignments, and discussion questions.

- Should you encounter any issues with the website, promptly inform the instructor.

COURSE READINGS

Software Manuals

- Qualtrics. https://www.qualtrics.com/support/survey-platform/survey-module/survey-module-overview/?parent=p0027
- IBM SPSS Statistics. https://www.ibm.com/docs/en/spss-statistics/29.0.0
- R. https://cran.r-project.org/doc/manuals/r-release/R-intro.pdf

Articles

- Shrout, P. E., & Bolger, N. (2002). Mediation in experimental and nonexperimental studies: New procedures and recommendations. *Psychological Methods, 7*, 422–445. https://doi.org/10.1037/1082-989X.7.4.422
- Hox, J. (2010). *Multilevel analysis: Techniques and applications* (2nd ed.), Erlbaum.

Supplemental Course Materials, including journal articles, book chapters, and videos, will be provided in electronic formats (such as PDFs and online links) on the course website.

ASSESSMENT OF LEARNING OUTCOMES & GRADING

- This course includes a set of assignments, each involving one or more of the following tasks: conducting a literature review, planning a research study, analyzing data, interpreting results, and reporting findings.
- By the end of the semester, you will be expected to develop your own research project and related products (e.g., project plan, results summary, manuscript draft). Throughout the semester, we will work on various components that serve as building blocks for your final project.
- Each written assignment will be evaluated on two key aspects:
    - **Quality of Work:** This includes the relevance of selected theories and techniques, technical precision, and so forth.
    - **Quality of Presentation:** This encompasses the neatness of the submission, clarity of expression, effective use of tables/graphics, adherence to APA publication guidelines, and more.

- Below is a detailed breakdown of the grading criteria and the final grade scale.

Assignments	Points	%
Module I: Online Survey using Qualtrics		
#1: Qualtrics Exercise	90	21 %
Module II: Intermediate Statistics using SPSS		
#2: Data Day	20	5 %
#3: Research Prospectus	90	21 %
Module III: Intermediate Statistics using R		
#4: Data Analysis & Report	160	38 %
#5: Revise & Resubmit	60	14 %
Module IV: Structural Regression & Mediation		
#6: (extra credit) Mediation Test	60	

A+	≥97 %	B+	87–89 %	C	70–79 %
A	93–96 %	B	83–86 %	D	60–69 %
A-	90–92 %	B-	80–82 %	F	≤59 %

ASSIGNMENT POLICY

- **Format Requirements:** All written assignments must follow APA publication guidelines. Submissions should be typed in a Word document with 1-inch margins on all sides, 12-point Times New Roman font, and double-spaced. Failure to adhere to these standards will result in points deduction.

- **Header Details:** Do not include a cover page. Instead, include your full name and the assignment number in the document's header (e.g., "Jane Doe Assignment #1").

- **Submission Deadline:** Each assignment must be submitted to the course website by 11:59 pm on the due date.

- **File Naming Convention:** The Word document should be named in the following format:

  "stats_soft -firstname_lastname-assignment.docx" (use all lowercase letters, and only include dashes and underscores; no special characters or spaces)

  Example: "stats_soft-jane_doe-assignment4.docx" (Word document file)
  "stats_soft-jane_doe-assignment4.R" (R script file)

- **Late Submission Policy:** Assignments submitted past the due date will not be accepted or graded. Ensure timely submission. Exceptions are made only for valid reasons communicated to the instructor in advance. Note: Computer failure is not a valid excuse for late submissions. In case of technical difficulties, email your assignment by the deadline.

For extraordinary circumstances (e.g., hospitalization, jury duty), please discuss with the instructor for potential makeup arrangements.

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COURSE SCHEDULE (due date)

Modules & Topics/activities	Class meetings	Assignments
Module I: Online Survey using Qualtrics		
Create an online survey Set up rules and functions Distribute the survey, send reminders	Jan 23, 12-4 pm (EDUC 300)	#1: Qualtrics Exercise (2/2)
Module II: Intermediate Statistics using SPSS		
Data manipulation Correlation, regression *t*-test, ANOVA Repeated measures	Jan 24, 8-12 am (EDUC 300)	#2: Data Day (2/3-9) #3: Research Prospectus (2/16)
Module III: Intermediate Statistics using R		
Basic functions Data manipulation using {tidyverse} Same topics as in Module II	Feb 20, 12-4 pm Feb 21, 8-12 am (EDUC 300)	#4: Data Analysis & Report (3/9) #5: Revise & Resubmit (3/23)
Module IV: Structural Regression & Mediation		
Structural regression Mediation Bootstrapping Reporting (tables, graphs)	March 27, 12-4 pm March 28, 8-12 am (EDUC 300)	#6: Mediation test (4/13)

Note. This schedule may be altered when needed, and some topics may be deleted or added with consideration of the content and students' interest in the topics.